

# Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference

---

Abhi Jain

August 4, 2026

PhD Candidate

Department of Biostatistics

Boston University School of Public Health

# Table of contents

1. Motivating problem
2. Causal inference with interference
3. Bayesian additive regression trees (BART)
4. Simulations
5. Case study: Effect of healthy food access on obesity rates in Texas
6. Discussion

# Motivating problem

---

# Effect of healthy food access on obesity rates

Poor healthy food access been associated with:

- high systolic blood pressure (Suarez et al., 2015)
- hypertension and cardiovascular disease (Kelli et al., 2019)
- obesity (Thomsen et al., 2016)

Challenges include:

- **Interference:** cross-border shopping
- **Heterogeneous effects:** magnitude and direction of effects can vary by community characteristics

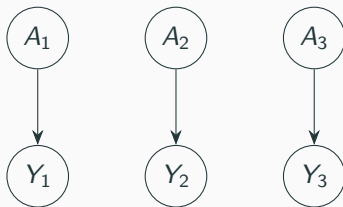
**Research question:** What is the effect of healthy food retailers on ZIP Code obesity rates and how does it vary for different ZIP Code characteristics?

# Causal inference with interference

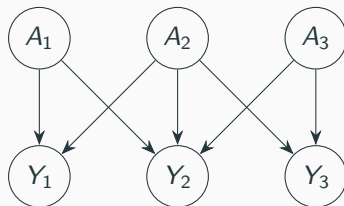
---

# Interference

- **No interference assumption:** The treatment/exposure of one unit does not impact the outcome of another unit
- Often violated in studies with spatial treatments or exposures



**Figure 1:** No interference



**Figure 2:** Spatial network interference

# Notation and Potential outcomes

- $\mathbf{W}$  = adjacency matrix
  - $w_{ij}$  = strength of connection between units  $i$  and  $j$
  - if  $w_{ij} = 0$ , units  $i$  and  $j$  are unconnected
- $\mathcal{N}_i$  = set of units that are connected to unit  $i$ 
  - Call this set the neighbors of unit  $i$
- $(i, \mathcal{N}_i, \mathcal{N}_{-i})$  = partition of all units
- Potential outcome:  $Y(A_i, \tilde{A}_i = f(\mathbf{A}_{\mathcal{N}_i}))$ 
  - $(A_i, \tilde{A}_i)$  = joint treatment
  - $f(\mathbf{A}_{\mathcal{N}_i})$  is a summarizing function of the neighboring exposures/treatments

- **Direct effect (DE):** effect of unit-level treatment ( $A$ ) on  $Y$  holding neighbor-level treatment fixed

$$DE(\tilde{A} = \tilde{a}) = E[Y(1, \tilde{a}) - Y(0, \tilde{a})]$$

- **Indirect (spillover) effect (IE):** effect of neighbor-level treatment ( $\tilde{A}$ ) holding unit-level treatment fixed

$$IE(A = a) = E[Y(a, 1) - Y(a, 0)]$$

- For simplicity, we consider binary  $A$  and  $\tilde{A}$

# Assumptions

1. Consistency:  $Y_i = Y_i(\mathbf{A})$
2. **Neighborhood interference**: Given a function  $f$ , if  $f(\mathbf{A}_{\mathcal{N}_i}) = f(\mathbf{A}'_{\mathcal{N}_i})$ , then  $Y(A_i, \mathbf{A}_{\mathcal{N}_i}) = Y(A_i, \mathbf{A}'_{\mathcal{N}_i})$  (Forastiere et al., 2021; Hudgens and Halloran, 2008)
3. Unconfoundedness of the joint treatment:  
 $Y_i(a, \tilde{a}) \perp A_i, \tilde{A}_i | \mathbf{X} \quad \forall a, \tilde{a}, \quad \forall i$
4. Positivity:  $f(a, \tilde{a} | \mathbf{X}) > 0 \quad \forall \mathbf{x} \quad \text{s.t.} \quad f_{\mathbf{X}}(\mathbf{x}) > 0$

## Bayesian additive regression trees (BART)

---

- Ensemble machine learning method that models the outcome as a sum of weak learners
- Bayesian framework: places priors on tree structure and parameters and provides uncertainty quantification via credible intervals
- Often used for causal inference problems (Hill, 2011; Dorie et al., 2022)

# Bayesian causal forests (BCF)

- Extension of BART designed for causal inference and heterogeneous treatment effects (Hahn et al., 2020)

$$Y_i = \mu(\mathbf{X}_i, \hat{\pi}(\mathbf{X}_i)) + \tau(\mathbf{X}_i)A_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- $\mu(\mathbf{X})$  = prognostic function
- $\tau(\mathbf{X})$  = moderating function
- Include estimate of the propensity score in the prognostic function

# Joint treatment Bayesian causal forests (jtBCF)

- Extension of Bayesian causal forests for joint treatments
- Separate moderating functions for both  $A$ ,  $\tilde{A}$ , and their interaction

$$Y_i = \mu(\mathbf{X}_i, \hat{\pi}_A, \hat{\pi}_{\tilde{A}}) + \tau_1(\mathbf{X}_i)A_i + \tau_2(\mathbf{X}_i)\tilde{A} + \tau_3(\mathbf{X}_i)A\tilde{A} + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$

- $\mu(\mathbf{X})$ : prognostic effect
- $\tau_1(\mathbf{X})$ : direct effect (own treatment)
- $\tau_2(\mathbf{X})$ : indirect (spillover) effect
- $\tau_3(\mathbf{X})$ : interaction effect

- We use the SoftBart R package (Linero, 2022) to implement our model
  - SoftBart R package also includes functionality to develop custom BART models
- Soft BART replaces binary splits with smooth gating functions (e.g., logistic functions) so each observation can partially belong to multiple leaves (Linero and Yang, 2018)

# Simulations

---

# Simulation setup

- Use Texas ZIP Codes as set of units ( $n = 1927$ )
- Simulate three unit-level covariates and compute three neighborhood-level covariates, which are neighborhood means of the unit-level covariates
- Consider linear/nonlinear  $\mu(\mathbf{X})$
- Consider homogeneous/heterogeneous  $\tau_1(X) - \tau_3(X)$
- Treatment:

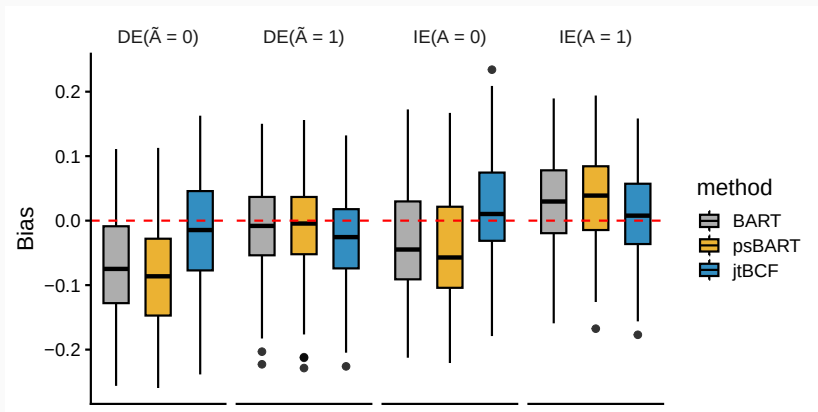
$$A \sim 0.8 \Phi(-1.25 + \mu(X)/\text{sd}(\mu(X)) + 0.5X_5) + u_i/10$$

$$\tilde{A}_i = \mathbf{1}(\bar{A}_{\mathcal{N}_i} > 0.5)$$

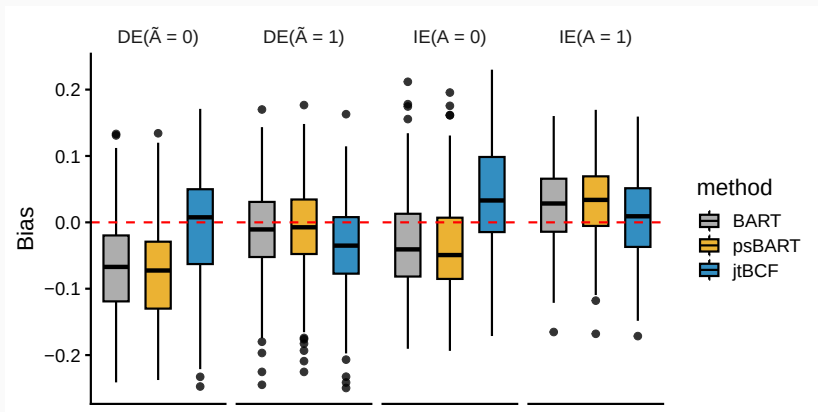
- Outcome:

$$y \sim N(\mu(X) + \tau_1(X)A + \tau_2(X)\tilde{A} + \tau_3(X)A\tilde{A}, 1)$$

## Simulation results (Linear/Homogeneous case)



# Simulation results (Linear/Heterogeneous case)



## Coverage - Direct and Indirect effects

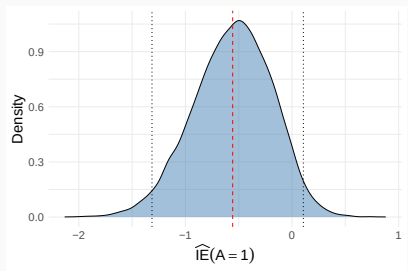
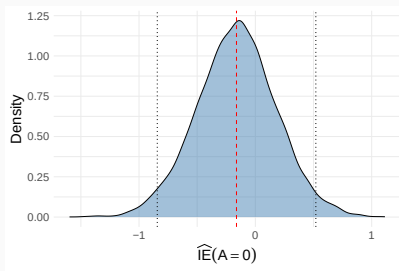
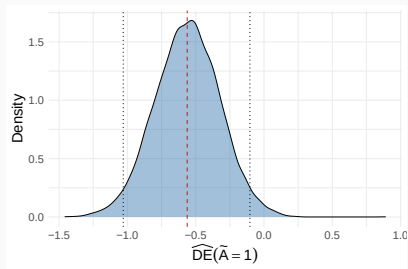
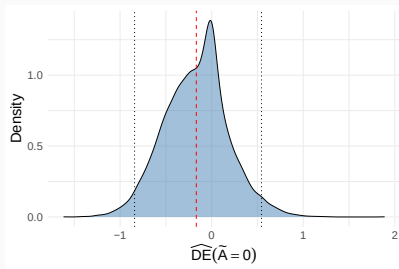
| Setting       | Estimand            | Soft BART | Soft psBART | Soft jtBCF |
|---------------|---------------------|-----------|-------------|------------|
| <b>Homo</b>   |                     |           |             |            |
|               | $DE(\tilde{A} = 0)$ | 0.80      | 0.78        | 0.94       |
|               | $DE(\tilde{A} = 1)$ | 0.94      | 0.94        | 0.93       |
|               | $IE(A = 0)$         | 0.90      | 0.86        | 0.93       |
|               | $IE(A = 1)$         | 0.91      | 0.90        | 0.96       |
| <b>Hetero</b> |                     |           |             |            |
|               | $DE(\tilde{A} = 0)$ | 0.82      | 0.82        | 0.96       |
|               | $DE(\tilde{A} = 1)$ | 0.92      | 0.93        | 0.93       |
|               | $IE(A = 0)$         | 0.93      | 0.93        | 0.92       |
|               | $IE(A = 1)$         | 0.95      | 0.95        | 0.97       |

## **Case study: Effect of healthy food access on obesity rates in Texas**

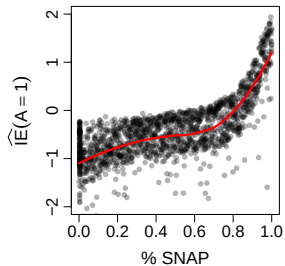
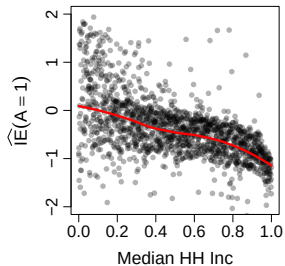
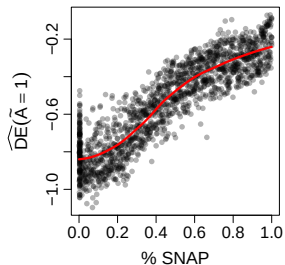
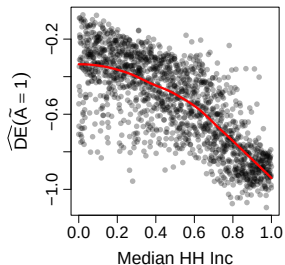
---

- **Unit of analysis:** Texas ZIP Codes in 2022
- **Outcome:** Obesity prevalence ( $y \in (0, 100)$ )
  - Model-based estimate adjusted for age, sex, race, and education
- **Exposures:**
  - $A$ : Indicator for  $\geq 1$  healthy food retailer in the ZIP Code.
  - $\tilde{A}$ : Indicator for  $\geq 1$  healthy food retailer in neighboring ZIP Codes.
- **Covariates:** Median household income, % SNAP, unemployment, % commuting by public transit, NDVI, % married, % veterans, and % with disability
- **Neighborhood:** Informed by driving times between ZIP Code centroids

# Posterior distributions of direct/indirect effects



# Results: Heterogeneous effects



# Discussion

---

# Conclusions

- jtBCF model offers bias reduction in estimating direct and indirect effects under interference compared to BART and psBART
- Provides strong coverage of direct and indirect effects
- Limitations:
  - Model has increased variability due to noise coming from three separate moderating functions
  - Motivating example has some weaknesses:
    - Data is ecological
    - Obesity rate at ZIP Code level is model-based estimate (takes into account race, sex, age, and education)
    - Food retailer list is not exhaustive; does not include farmer's markets or food banks

# Thank you!



Shariq Mohammed



Sara Lodi



Wei Jin



QR code for slides

## Any questions?

# References

---

- Vincent Dorie, George Perrett, Jennifer L Hill, and Benjamin Goodrich. Stan and bart for causal inference: Estimating heterogeneous treatment effects using the power of stan and the flexibility of machine learning. *Entropy*, 24(12):1782, 2022.
- Laura Forastiere, Edoardo M Airoidi, and Fabrizia Mealli. Identification and estimation of treatment and interference effects in observational studies on networks. *Journal of the American Statistical Association*, 116(534):901–918, 2021.
- P Richard Hahn, Jared S Murray, and Carlos M Carvalho. Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*, 15(3):965–1056, 2020.
- Jennifer L Hill. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011.
- Michael G Hudgens and M Elizabeth Halloran. Toward causal inference with interference. *Journal of the american statistical association*, 103(482):832–842, 2008.

- Heval M Kelli, Jeong Hwan Kim, Ayman Samman Tahhan, Chang Liu, Yi-An Ko, Muhammad Hammadah, Samaah Sullivan, Pratik Sandesara, Ayman A Alkhoder, Fahad K Choudhary, et al. Living in food deserts and adverse cardiovascular outcomes in patients with cardiovascular disease. *Journal of the American Heart Association*, 8(4):e010694, 2019.
- Bora Kim. Estimating spillover effects in the presence of isolated nodes. *Spatial Economic Analysis*, pages 1–15, 2025.
- Antonio R Linero. Softbart: Soft bayesian additive regression trees. *arXiv preprint arXiv:2210.16375*, 2022.
- Antonio R Linero and Yun Yang. Bayesian regression tree ensembles that adapt to smoothness and sparsity. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 80(5):1087–1110, 2018.
- Jonathan J Suarez, Tamara Isakova, Cheryl AM Anderson, L Ebony Boulware, Myles Wolf, and Julia J Scialla. Food access, chronic kidney disease, and hypertension in the us. *American journal of preventive medicine*, 49(6):912–920, 2015.
- Michael R Thomsen, Rodolfo M Nayga Jr, Pedro A Alviola IV, and Heather L Rouse. The effect of food deserts on the body mass index of elementary schoolchildren. *American Journal of Agricultural Economics*, 98(1):1–18, 2016.

$$\mu(\mathbf{x}) = \begin{cases} 1 + 0.7X_1 - 0.6X_2 + X_3 + 0.3X_4, & \text{linear} \\ 0.5 + 0.6|X_1 + 1| - 0.8\sin(\pi X_2) + X_3 - 0.6X_4 + 0.3X_1X_4, & \text{nonlinear} \end{cases}$$

$$\tau(\mathbf{x}) = \begin{cases} 0.4A + 0.4\tilde{A} + 0.2A\tilde{A}, & \text{homo} \\ 0.3(1 + \frac{X_1}{2} + X_3)A + 0.3(1 + \frac{X_2}{2} + X_3)\tilde{A} + 0.15(X_3 + X_6)A\tilde{A}, & \text{hetero} \end{cases}$$

## Results: Overall effects

- $DE(\tilde{A} = 0) = -0.17(-0.84, 0.54)$
- $DE(\tilde{A} = 1) = -0.56(-1.03, -0.10)$
- $IE(A = 0) = -0.16(-0.86, 0.50)$
- $IE(A = 1) = -0.56(-1.25, 0.17)$

# Isolated nodes

- Since we use driving times to inform the neighborhood adjacency, some ZIP Codes are isolated (no neighbors)
- Often it is assumed  $\tilde{A} = 0$  for isolated nodes
- We drop these ZIP Codes from the analysis since it has been shown that estimation of spillover effects is biased when we assume  $\tilde{A} = 0$  for isolated nodes (Kim, 2025)